

Azure Lab Setup #3

Introduction

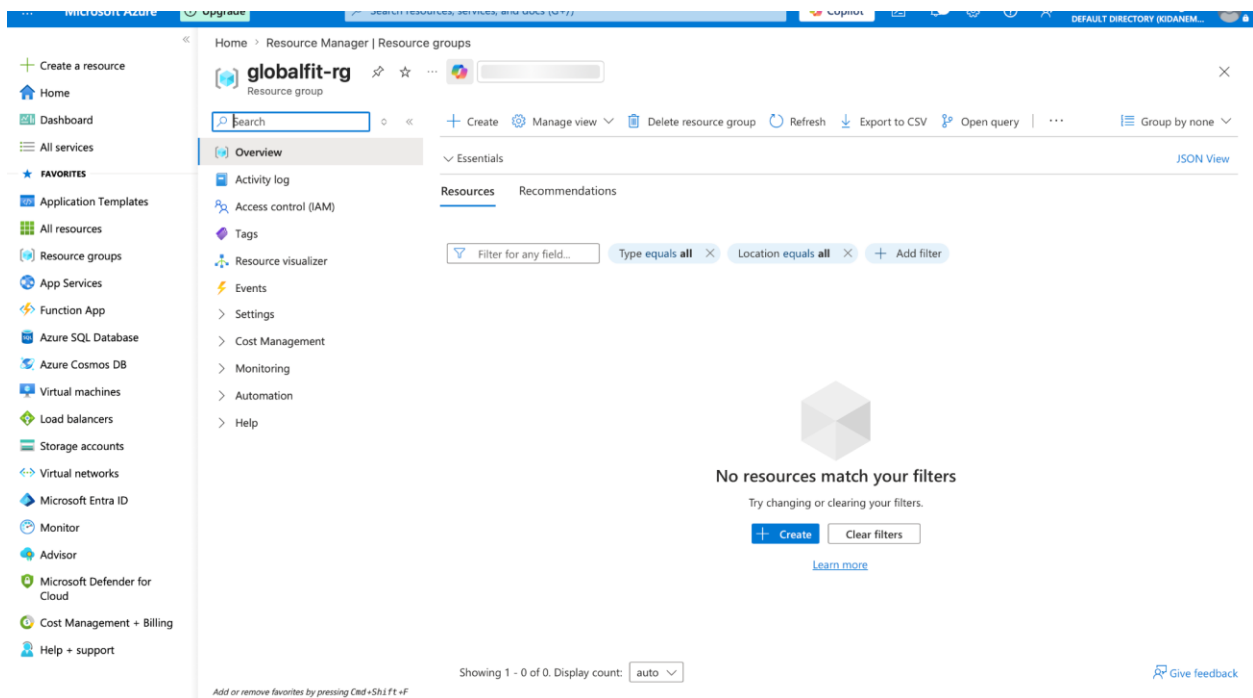
ChartPath Medical is a cloud based electronic medical records company that supports private clinics and outpatient practices with secure patient charting, lab results, and appointment data. The company needed a full Azure environment to host its application, store patient files, run a NoSQL database for visit records, and give two subsidiary clinic groups controlled access to the main organization's resources.

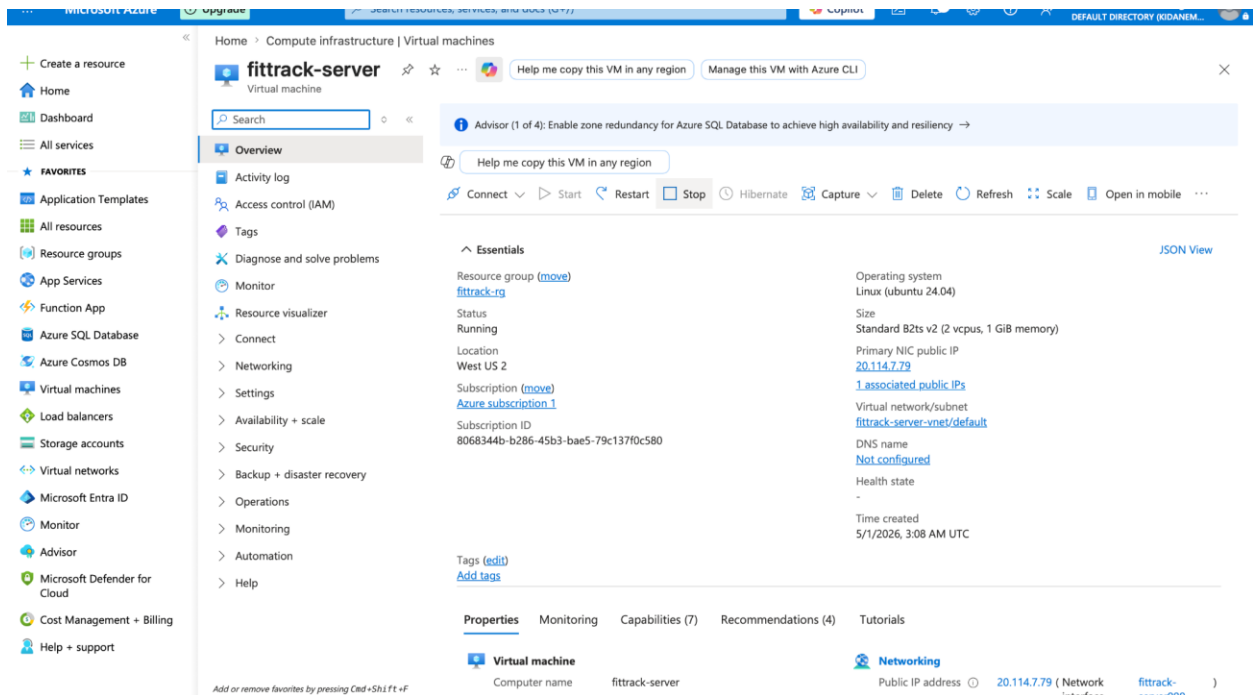
Business Requirements

- Geo redundant replication for patient data
- Cosmos DB with continuous backups and snapshots
- Blob storage for clinic uploaded files
- Archive tier storage with restricted permissions for older records
- Encryption for all stored and transferred data
- Identity and Access Management for the two subsidiary clinics
- Performance and uptime monitoring
- A hosting server for the EMR application

Server

The existing virtual machine fittrack-server was reused as the main application server for ChartPath. It handles backend traffic for the EMR portal including authentication, chart updates, and report generation. A separate VM was also configured to act as the database connection layer between the application and Cosmos DB.





Database

Azure Cosmos DB was deployed using the SQL API to store visit notes and patient encounter records as JSON documents. Continuous backup was enabled so any deletion or corruption could be recovered to a specific point in time. Manual snapshots were also documented as part of the recovery plan.

Create Azure Cosmos DB Account - Azure Cosmos DB for MongoDB

Validation errors were found. Click here for details →

Basics Global distribution Networking Backup Policy Security Tags Review + create

Creation Time

Estimated Account Creation Time (in minutes) 6

The estimated creation time is calculated based on the location.

Basics

Subscription	Azure subscription 1
Resource Group	globalfit-rg
Location	Central US EUAP
Account Name	(new) globalfit-cosmos123
API	Azure Cosmos DB for NoSQL
Capacity mode	Serverless

Backup Policy

Backup policy	Periodic
Backup storage redundancy	Locally-redundant backup storage

Security

Key-based Authentication (new) Enable

Create Previous Next Download a template for automation

Errors

Summary Raw Error

Copy Full Error

TOP ERROR

The provided location 'centraluseap' is not available for resource type 'Microsoft.DocumentDB/databaseAccounts'. List of available regions for the resource type is 'westcentralus,northcentralus,centralus,brazilsouth,canadacentral,westus2,eastus2,francecentral,japaneast,southeastasia,centralindia,southafricanorth,switzerlandnorth,uenaorth,ukwest,northeurope,westeurope,norwayeast,koreacentral,austrialaest,canadaeast,eastasia,germanywestcentral,uksouth,austrialacentral,austriasoutheast,japanwest,koreasouth,southindia,westindia,westus,southcentralus,eastus,westus3,jioindiawest,swedencentral,qatarcentral,polandcentral,itainorth,israelcentral,spaincentral,newzealandnorth,mexicoentral,indonesiacentral,chilecentral,malaysiawest,austrialaest,belgiumcentral,demmarkeast'. (Code: LocationNotAvailableForResourceType)

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Troubleshooting Options

New Support Request 📄

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A storage account was created to hold clinical files such as scanned intake forms, x-ray images, and lab PDFs. A container named patient-files was used to keep clinical uploads separate from administrative documents. Access was scoped using shared access signatures so subsidiary users only see the containers their clinic owns.

Microsoft Azure | Upgrade | DEPLOYMENT RESOURCES, SERVICES, BING MAPS (10/7/7) | Copilot | DEFAULT DIRECTORY (KIDANEM...)

Home

globalfitstorage1233_1777958110401 | Overview

Deployment

Search [] X << [] Delete [] Cancel [] Redeploy [] Download [] Refresh []

Overview

Your deployment is complete

Deployment name : globalfitstorage1233_17779... Start time : 5/4/2026, 10:15:25 PM
 Subscription : Azure subscription 1 Correlation ID : d72beff0-37d2-4efa-9428-2...
 Resource group : globalfit-rg

Deployment details

Resource	Type	Status	Ops
globalfitstorage123	Microsoft.Storage/storageAccou	OK	Ops
globalfitstorage123	Microsoft.Storage/storageAccou	OK	Ops
globalfitstorage123	Storage account	OK	Ops

Next steps

[Go to resource](#)

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Home > Resource Manager | All resources > globalfitstorage1233

globalfitstorage1233 | Containers

Storage account

Search [] X << [] Add container [] Upload [] Refresh [] Delete [] Change access level [] Restore containers [] Edit columns []

Search containers by prefix [] Only show active containers []

Showing all 2 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	logs	5/4/2026, 10:15:49 PM	Private	Available	-
☐	company-files	5/4/2026, 10:19:01 PM	Private	Available	-

Containers

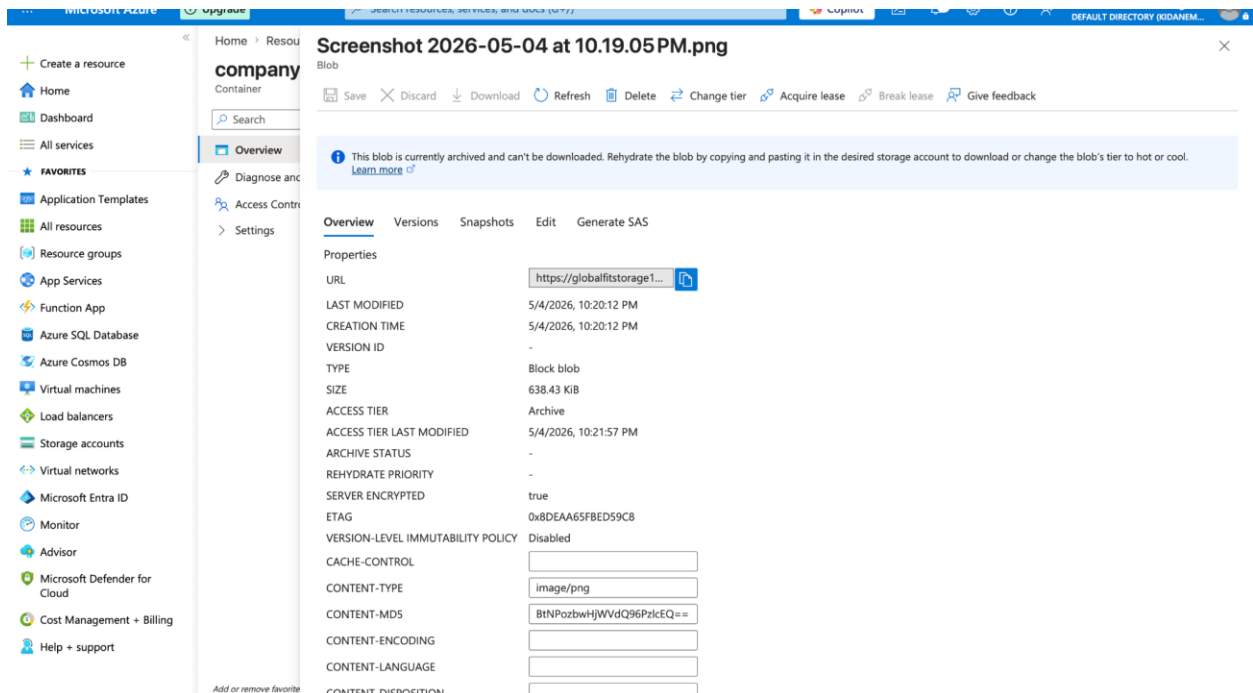
File shares
Queues
Tables

> Security + networking
 > Data management
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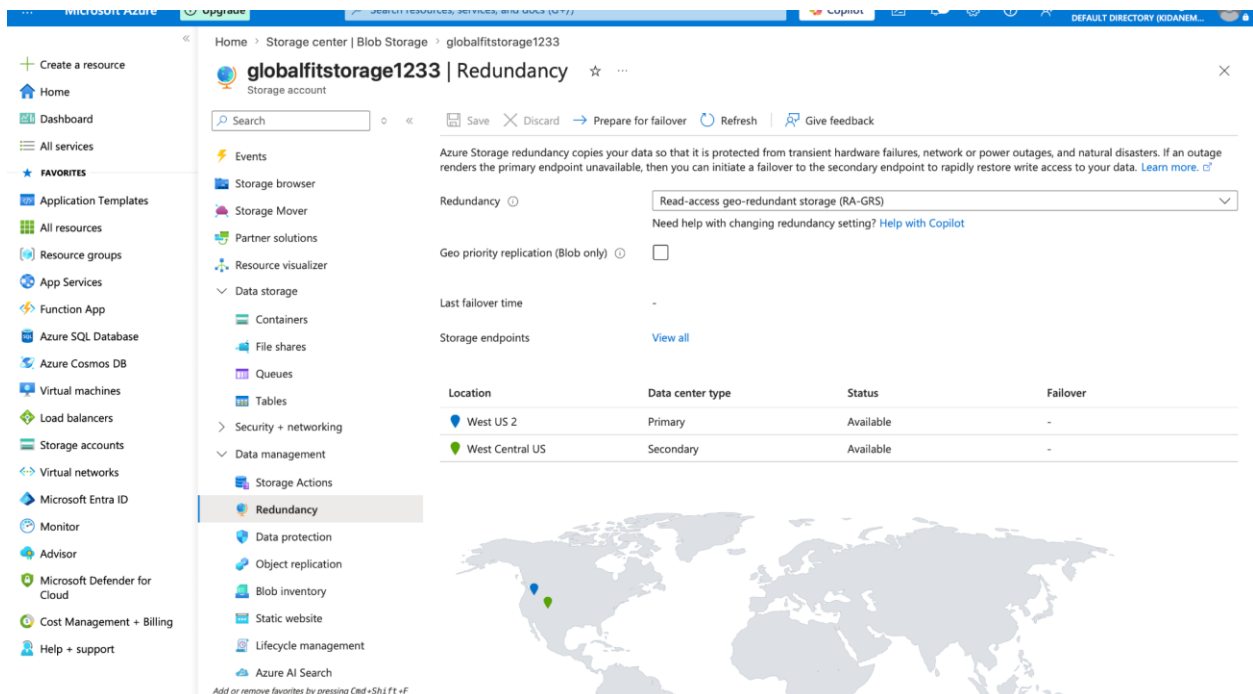
Archive Storage

Files older than five years were moved to the Archive access tier to lower storage cost. Permissions on the archive container were restricted to compliance staff only, since archived patient records still need to meet HIPAA retention rules.



Replication

Geo redundant storage was enabled on the storage account so that all blob data is automatically replicated to a paired Azure region. This protects clinical data if the primary region goes down.



Identity and Access Management

Two subsidiary clinic groups were added to the environment, Pacific Coast Clinics and Mountain View Clinics. A user account was created for each clinic with role based access control. Each subsidiary user was assigned a reader role on its own container only, so the two clinics cannot see each other's files. A main administrator account holds Owner rights on the resource group.

Monitoring

Azure Monitor was set up to track VM performance, Cosmos DB request units, and storage account latency. Alerts were configured to email the admin when CPU stays above 80 percent for more than five minutes.

Encryption and Security

All Azure storage uses encryption at rest by default with platform managed keys. Data in transit is protected through TLS. Network security rules were added to block public access to the database and only allow connections from the main server.

Conclusion

The Azure environment satisfies the needs of ChartPath Medical by giving it a secure, replicated, and monitored platform for storing patient records while keeping the two subsidiary clinics inside separate access boundaries.